PADHAKU
ICT – Class 6 · Class 6 ICT



CHAPTER 5

Learn to Code-01: Learning to Create Flowchart and Programs

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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⭐ Practice

Practice 1 **Write an algorithm to calculate the area of a rectangle.**

✔ ANSWER

Algorithm to calculate the area of a rectangle

START

Step 1: Take the length ‘L’ of the rectangle.

Step 2: Take the breadth ‘B’ of the rectangle.

Step 3: Calculate L*B and name this value ‘A’ (Area = length × breadth).

Step 4: Print ‘A’.

STOP

Practice 2 **Make a flowchart to calculate the area of a rectangle (use Google Drawing).**

💡 SAMPLE ANSWER

Draw these shapes from top to bottom and join them with arrows (↓):

Shape	What to write inside
Oval	**Start**
Parallelogram (input)	Input length L and breadth B
Rectangle (process)	A = L × B
Parallelogram (output)	Print A
Oval	**Stop**

Then download the flowchart (File → Download) and open the file to check it.

Practice 3 **Draw the triangle shown in the book using Scratch.**

💡 SAMPLE ANSWER

- Choose a sprite and add the **Pen** extension.
- Script: **set pen size to 2** → **pen down** →
- **move 100 steps** → **turn 120 degrees** (3 times) →
- **pen up**. Double-click the script to see the output.

Turning 120° three times (3 × 120° = 360°) brings the sprite back to the start and makes a triangle with three equal sides. (check the picture in your book; change the steps and angle to match it)

❓ Exercises · Multiple Choice Questions

1 **Why is a flowchart used in computer programming?**

a. To show an algorithm in pictures (symbols)

b. To write a program

c. To find errors in a program

d. To design an interface for user input

✔ ANSWER

a. To show an algorithm in pictures (symbols)

A flowchart shows the steps of an algorithm as a diagram.

2 **What is an algorithm in computer programming?**

a. A set of instructions to solve a problem

b. A flowchart to represent code

c. A programming language

d. A debugging tool

✔ ANSWER

a. A set of instructions to solve a problem

An algorithm is the sequence of instructions needed to complete a task.

3 **What is the purpose of using Scratch in computer programming?**

a. To make interactive animations and games

b. A flowchart to represent code

c. To find errors in a program

d. To design an interface for user input

✔ ANSWER

a. To make interactive animations and games

Scratch is a block-based language for making animations, games and stories.

4 **What is the purpose of Motion blocks in Scratch?**

a. To control the movement of sprites on the stage

b. To change the colour of the stage background

c. To add sound effects to the project

d. To add text on the stage

✔ ANSWER

a. To control the movement of sprites on the stage

Motion blocks (move, turn) control how a sprite moves.

5 **What is the purpose of Pen blocks in Scratch?**

a. To add sound effects to the project

b. To control the movement of sprites on the stage

c. To draw lines and shapes on the stage as sprites move

d. To change the colour of the stage background

✔ ANSWER

c. To draw lines and shapes on the stage as sprites move

Pen blocks (pen down, pen up, set pen size) control the pen used for drawing.

6 What is the definition of a programming language?

a. A set of instructions to solve a problem

b. A special software application

c. A language used by humans in which written instructions can be carried out by a computer

d. A debugging tool

✔ ANSWER

c. A language used by humans in which written instructions can be carried out by a computer

Computer programs are written in this special kind of language.

7 What is the definition of control of flow in computer programming?

a. The order in which instructions in a program are carried out

b. The process of debugging errors in a program

c. The process of designing an interface for user input

d. The process of representing code in a flowchart

✔ ANSWER

a. The order in which instructions in a program are carried out

Control flow is the order in which a computer runs/executes the instructions.

❓ Exercises · Short Answer Questions

1 **What is the difference between an algorithm and a flowchart?**

✔ ANSWER

Algorithm	Flowchart
A sequence of instructions needed to complete a task.	A diagram (graph) that shows the steps of an algorithm.
Written in words, step by step.	Drawn with shapes (oval, parallelogram, rectangle) joined by arrows.
Made first.	Made from the algorithm to show it in pictures.

2 **Define programming language.**

✔ ANSWER

A programming language is a special kind of language in which computer programs are written.

Examples: Logo, Scratch, C++, Java and Python.

3 **What is the concept of Control Flow in computer programming?**

✔ ANSWER

Control flow is the order in which a computer runs/executes the instructions of a program.

It is of three types: **Sequence control**, **Selection control** and **Loop control**.

By default, instructions run in sequence, one after another from top to bottom (Sequence flow).

❓ Exercises · Long Answer Question

1 **Draw the shape shown in the book using Scratch.**

💡 SAMPLE ANSWER

Steps (for any shape):

- Choose a sprite and add the **Pen** extension (Add Extension button, bottom-left).
- Add **set pen size** and **pen down** from the Pen section.
- For each side, add **move (length) steps** and **turn (angle) degrees** from the Motion section.
- Add **pen up** at the end and double-click the script to see the output.

Example, square: set pen size to 2 → pen down → (move 100 steps, turn 90 degrees) × 4 → pen up.

Example, rectangle: (move 150, turn 90, move 80, turn 90) × 2. Triangle: (move 100, turn 120) × 3.

(check the picture in your book and use the matching sides and angles)

🎉 **Chapter 5 done! Revise the answers once more tomorrow.**